



**ALLIANCE
UNIVERSITY**
Private University established in Karnataka State by Act No. 24 of year 2010
Recognized by the University Grants Commission (UGC), New Delhi.

**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: Importing business datasets from flat files & MySQL and connecting with any other sources into Power BI

Submitted By:

Name: Bijjam Sharvani

Reg. No.:2411021240060

Date:22/8/26

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Experiment No.: 1

Gith hub link: <https://github.com/sharvani801/Business-Analytics-Projects>

Importing Business Datasets from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

Problem Statement

You are a Data Analyst in a retail company. Sales data is stored in Excel files, customer data in CSV files, and product information in another source. Management wants a centralized reporting system. Import data from multiple sources into Power BI, establish the required connections, and prepare the data for business reporting.

Steps Followed

Step 1: Data Collection

- Obtained Sales Data from Excel files.
- Obtained Customer Data from CSV files.
- Obtained Product Data from My SQL Database.

Step 2: Data Import

- Opened Power BI Desktop.
- Imported all datasets using the **Get Data** option.
- Loaded Excel, CSV files and My SQL data into Power BI.

Step 3: Data Cleaning and Transformation

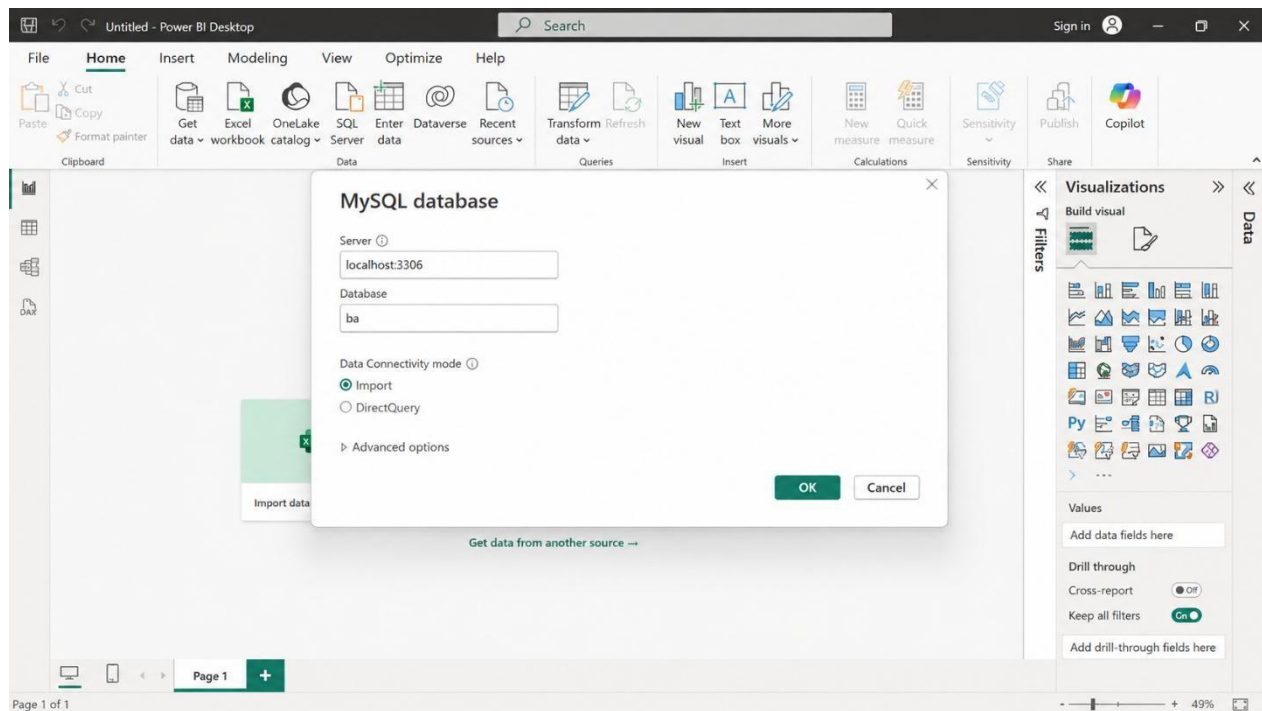
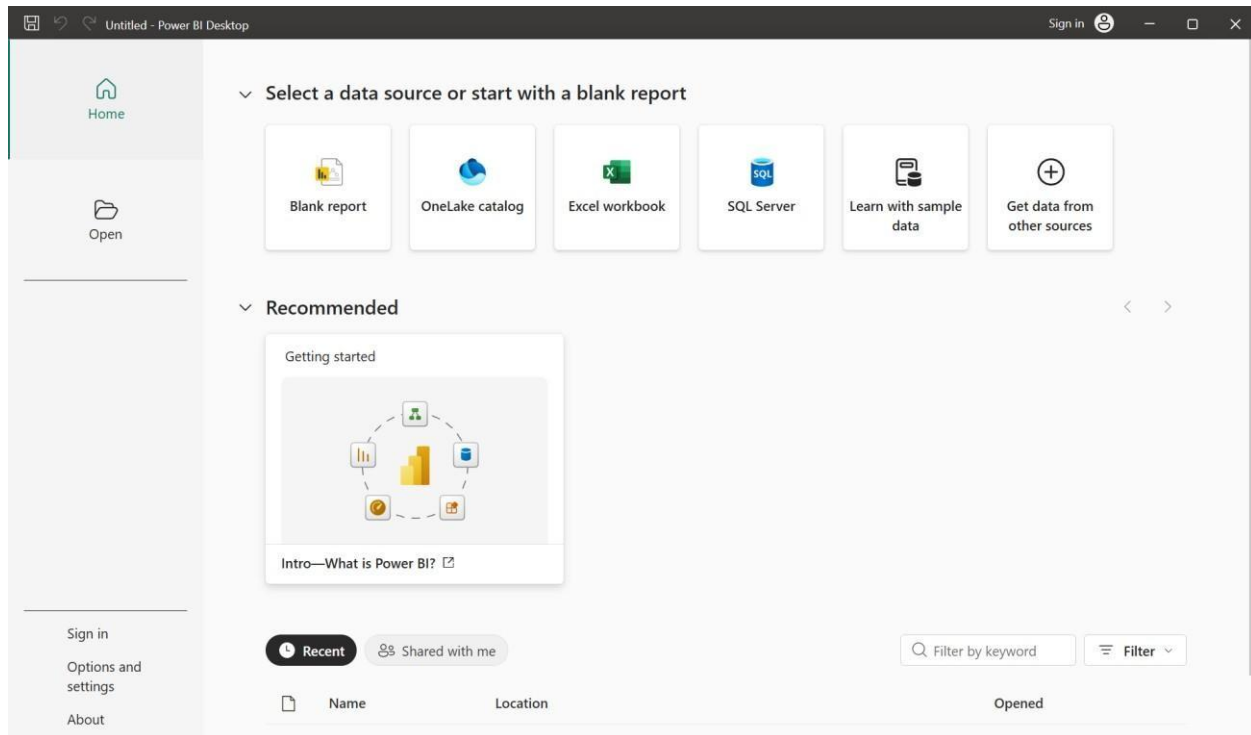
- Removed duplicate records.
- Handled missing values.
- Renamed columns appropriately.
- Converted data types where necessary.

Step 4: Data Modeling

- Created relationships between Sales, Customer, and Product tables.
- Verified one-to-many relationships.
- Created KPI cards for Total Sales, Orders, and Customers.
- Developed charts for Sales by Category and Region.
- Added slicers for Month, Category, and Region.
- Designed an interactive business dashboard.

Step 5: Dashboard Development

Output



Superstore - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Data source settings Manage Parameters Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Text Use First Row as Headers Replace Values Combine

Queries [3] Orders Returns People

12 123 Row ID Order ID Order Date Ship Date Ship Mode Customer ID

1	1	CA-2016-152156	11-11-2016	13-11-2016	Second Class	CG-12520
2	2	CA-2016-152156	11-11-2016	13-11-2016	Second Class	CG-12520
3	3	CA-2016-138688	12-06-2016	16-06-2016	Second Class	DV-13045
4	4	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335
5	5	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335
6	6	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710
7	7	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710
8	8	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710
9	9	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710
10	10	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710
11	11	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710
12	12	CA-2017-114412	15-04-2017	20-04-2017	Standard Class	AA-10480
13	13	CA-2017-114412	15-04-2017	20-04-2017	Standard Class	AA-10480
14	14	CA-2017-114412	15-04-2017	20-04-2017	Standard Class	AA-10480
15	15	CA-2017-114412	15-04-2017	20-04-2017	Standard Class	AA-10480
16	16	CA-2017-114412	15-04-2017	20-04-2017	Standard Class	AA-10480
17	17	CA-2017-114412	15-04-2017	20-04-2017	Standard Class	AA-10480
18	18	CA-2016-161389	05-12-2016	09-12-2016	Standard Class	IM-15070
19	19	CA-2016-161389	05-12-2016	09-12-2016	Standard Class	IM-15070
20	20	CA-2016-161389	05-12-2016	09-12-2016	Standard Class	IM-15070
21	21	CA-2016-161389	05-12-2016	09-12-2016	Standard Class	IM-15070
22	22	CA-2016-161389	05-12-2016	09-12-2016	Standard Class	IM-15070

10 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 10:45 AM

Query Settings

PROPERTIES

Name

Orders

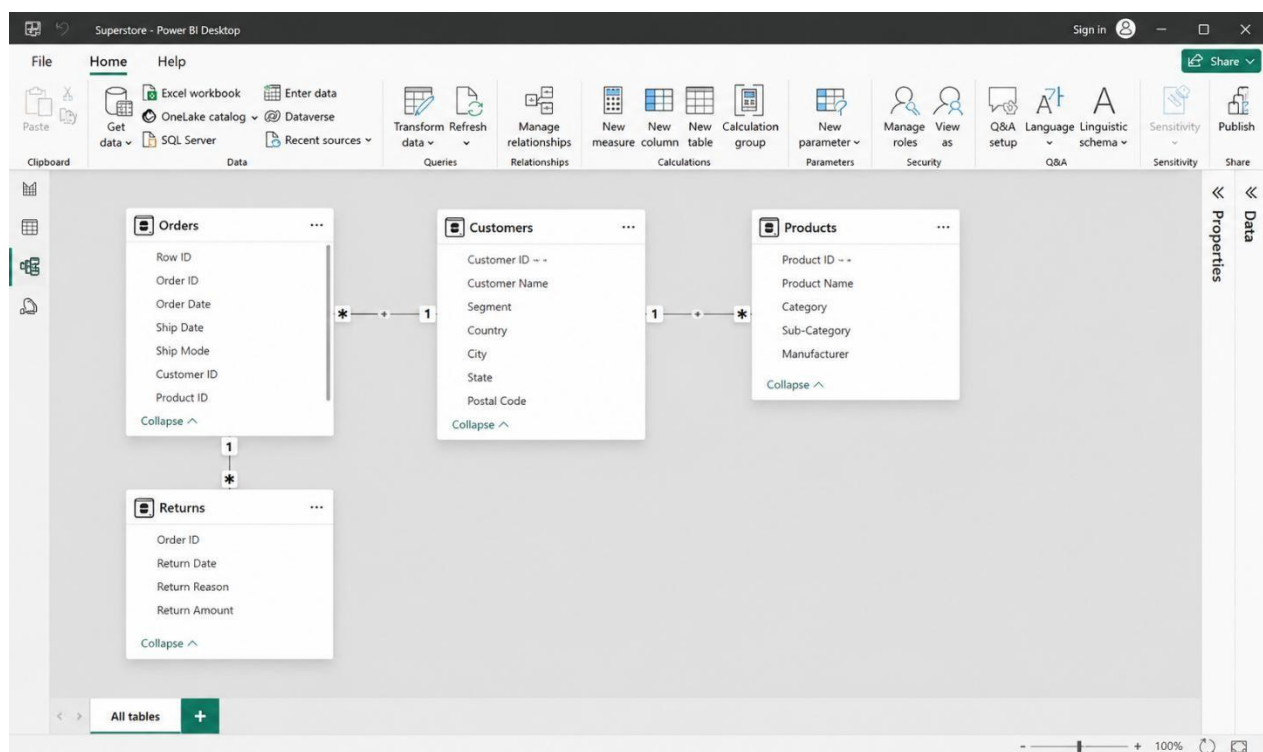
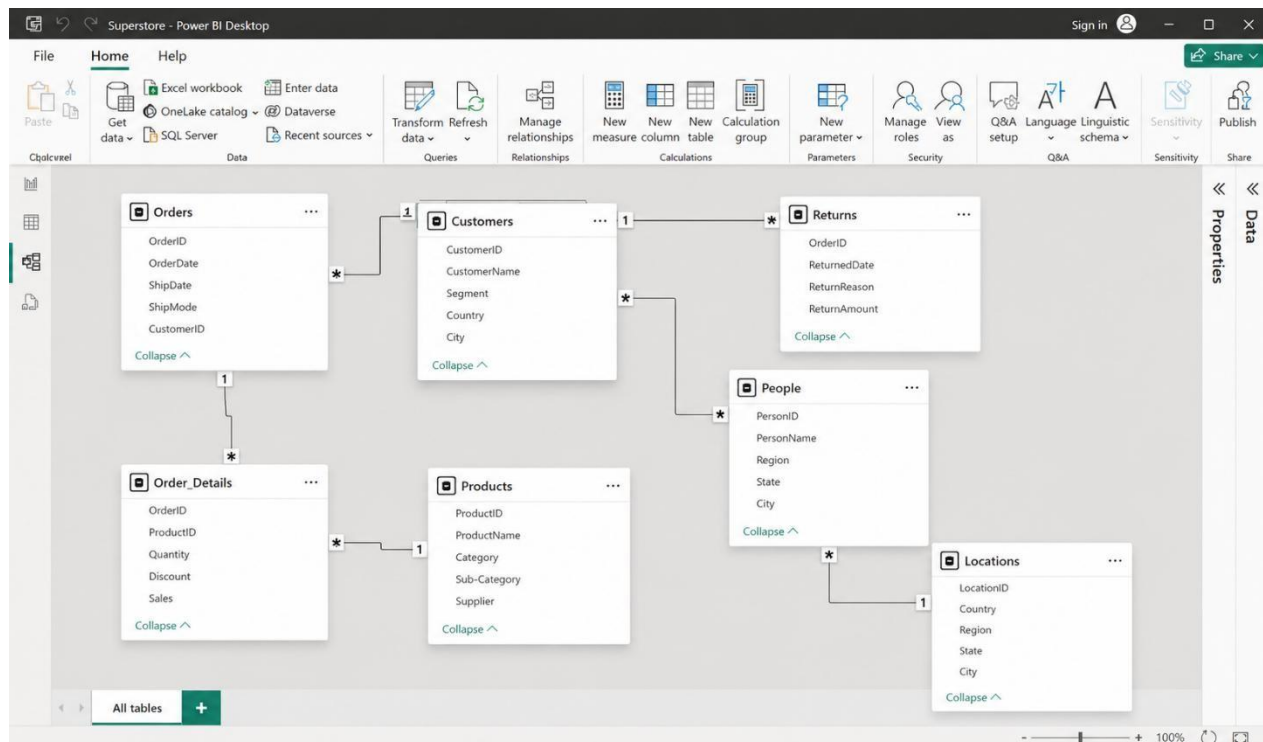
APPLIED STEPS

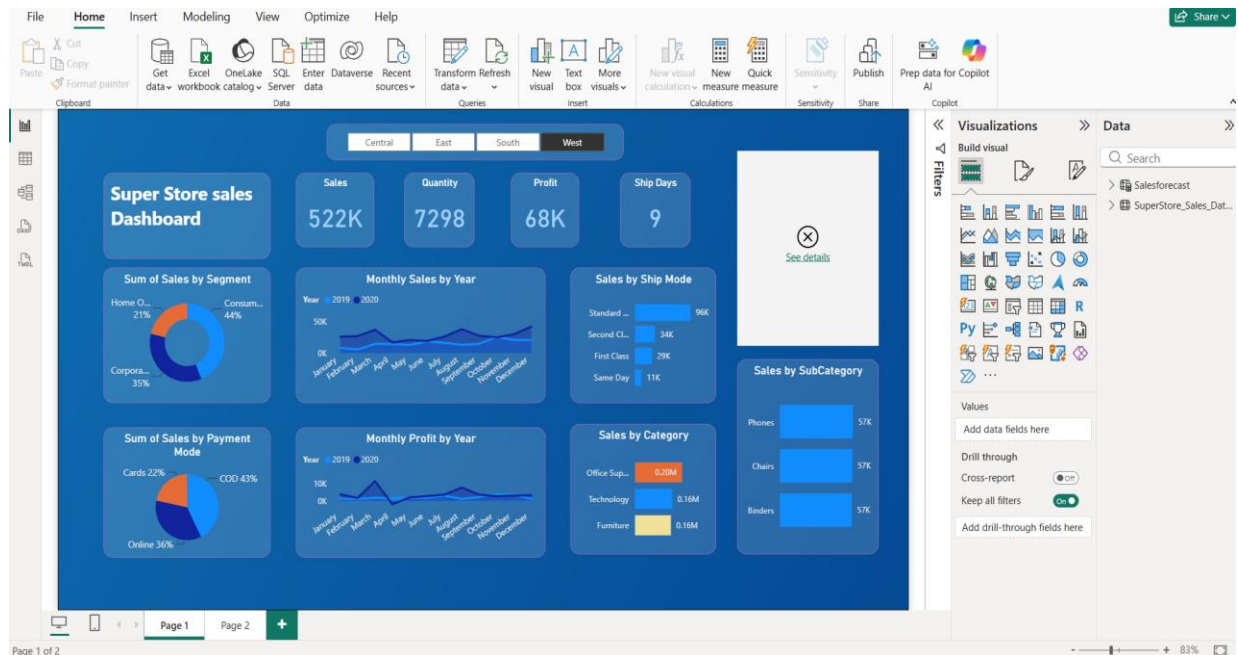
Source

Navigation

Promoted Headers

Changed Type





Final Power BI Dashboard – Superstore Dataset The

final Power BI dashboard provides:

- Total Sales and Profit Performance
- Customer-wise Sales Analysis
- Product and Sub-Category Sales Analysis
- Regional and State-wise Sales Performance
- Monthly and Year-wise Sales Trends
- Sales Analysis by Category and Segment
- Order and Quantity Analysis
- Shipping Mode Analysis
- Interactive Filters and Slicers for Region, Category, Sub-Category, Segment, and Order Date

The dashboard enables management to monitor sales, profit, customers, products, regions, and shipping performance and make effective data-driven business decisions using the Superstore dataset.

Conclusion

This project successfully integrated data from multiple sources, including Excel, CSV, and MySQL, into Power BI and transformed the raw data into meaningful business insights. Data cleaning and transformation techniques were applied to improve data quality, including checking data types, handling missing values, and ensuring consistency across the datasets.

The Sales, Customers, and Products tables were connected using appropriate relationships, creating a structured data model for analysis. The developed Power BI dashboard provides a centralized view of business performance through key metrics and visualizations such as total sales, orders, customer information, category-wise performance, regional sales, and monthly sales trends.

Overall, this project provided practical knowledge of data integration, data preparation, data modeling, and business intelligence using Power BI. The resulting dashboard makes the data easier to understand and supports effective analysis and better data-driven decision-making.

Learning Outcomes

Through this project, I learned:

- Understand how to import and integrate data from multiple sources, including Excel, CSV, and MySQL, into Power BI.
- Learn how to clean and transform raw data by changing data types, handling missing values, and checking for duplicate records.
- Understand the importance of data modeling and create appropriate relationships between Sales, Customers, and Products tables.
- Gain practical knowledge of creating DAX measures such as Total Sales, Total Orders, and Total Customers.
- Learn how to create and use interactive visualizations, charts, KPIs, and slicers in Power BI.
- Develop the ability to analyze sales performance, customer information, product categories, regional performance, and monthly trends.
- Understand how business data can be transformed into meaningful insights to support data-driven decision-making.
- Gain practical experience in using Power BI as a business intelligence and data visualization tool.